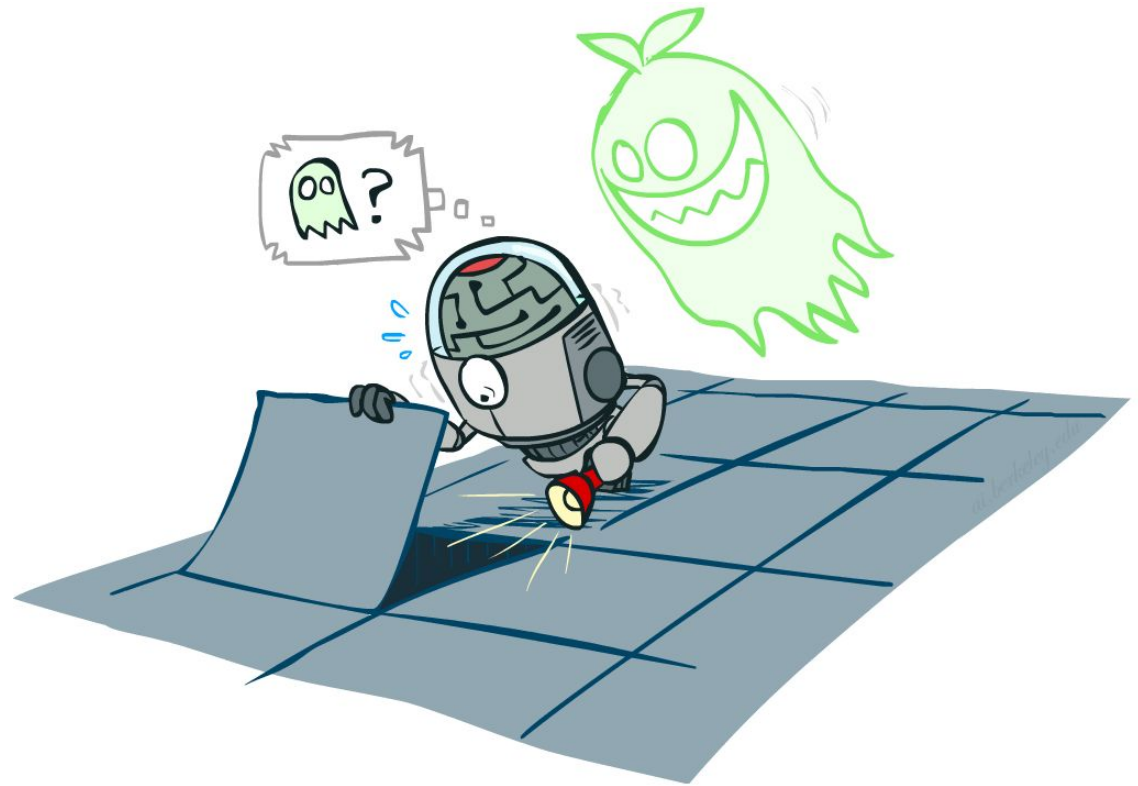


- We're done with Part I Search and Planning!

- Part II: Probabilistic Reasoning

- Diagnosis
- Speech recognition
- Tracking objects
- Robot mapping
- Genetics
- Error correcting codes
- ... lots more!

- Part III: Machine Learning



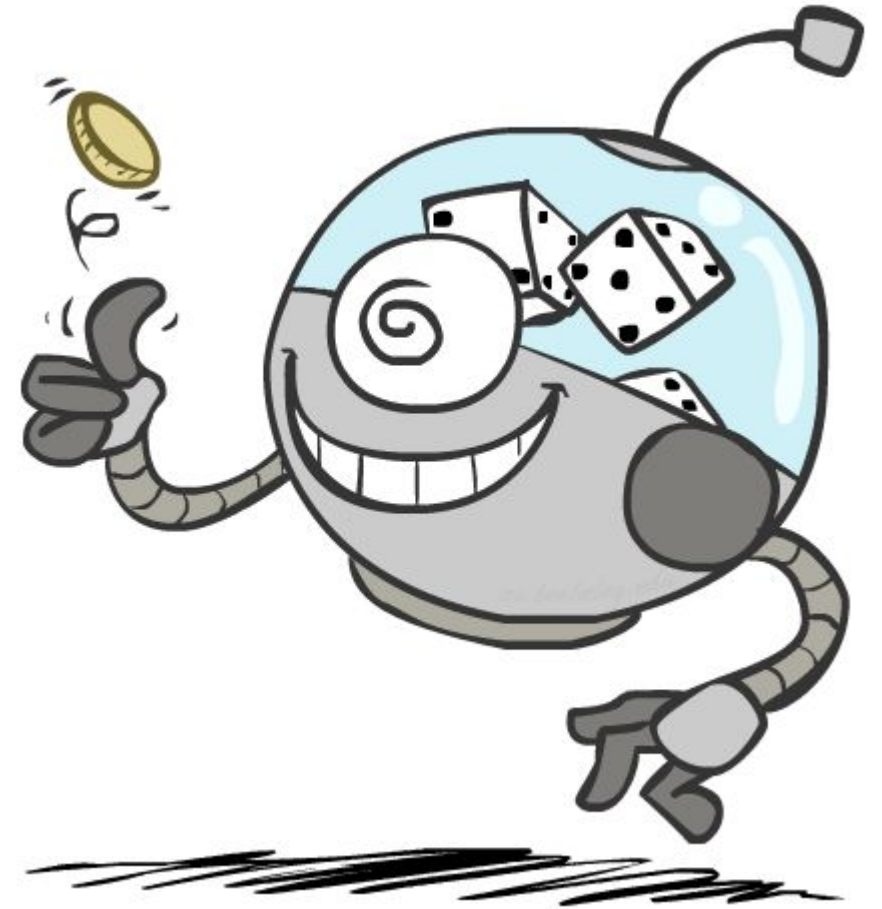
Artificial Intelligence

Probability



Today

- Probability
 - Random Variables
 - Joint and Marginal Distributions
 - Conditional Distribution
 - Product Rule, Chain Rule, Bayes' Rule
 - Inference
 - Independence



Uncertainty

- General situation:
 - **Observed variables (evidence):** Agent knows certain things about the state of the world (e.g., sensor readings or symptoms)
 - **Unobserved variables:** Agent needs to reason about other aspects (e.g. where an object is or what disease is present)
 - **Model:** Agent knows something about how the known variables relate to the unknown variables
- Probabilistic reasoning gives us a framework for managing our beliefs and knowledge

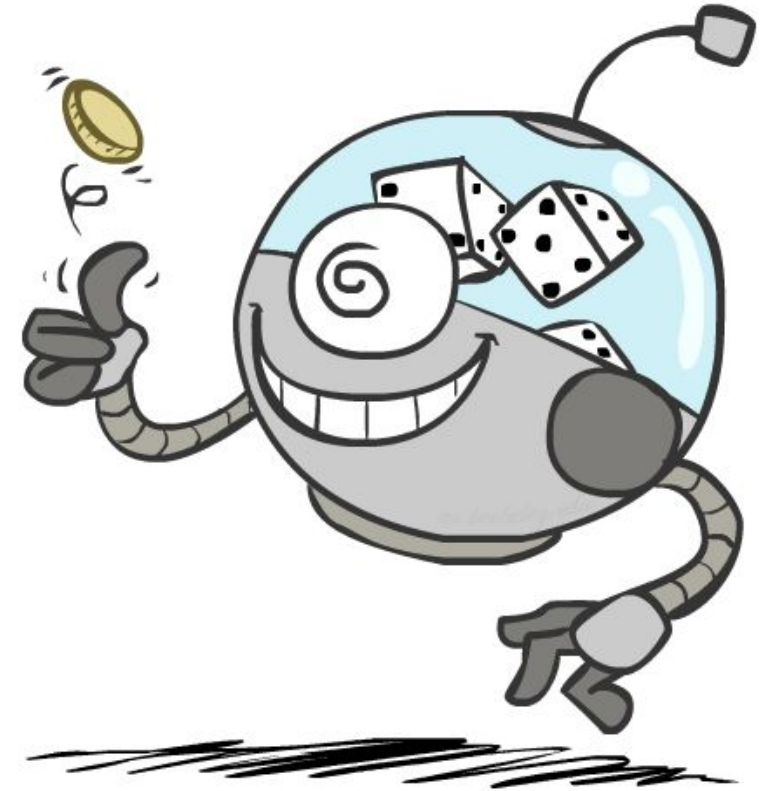
0.11	0.11	0.11
0.11	0.11	0.11
0.11	0.11	0.11

0.17	0.10	0.10
0.09	0.17	0.10
<0.01	0.09	0.17

<0.01	<0.01	0.03
<0.01	0.05	0.05
<0.01	0.05	0.81

Random Variables

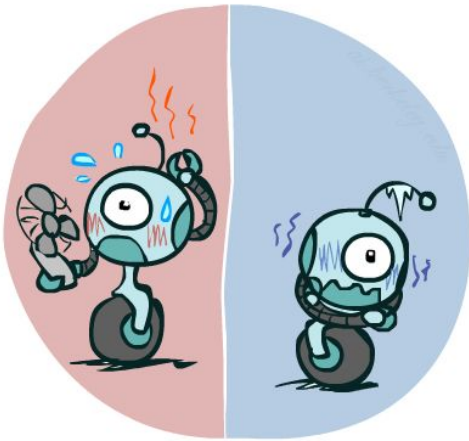
- A random variable is some aspect of the world about which we (may) have uncertainty
 - R = Is it raining?
 - T = Is it hot or cold?
 - D = How long will it take to drive to work?
 - L = Where is the ghost?
- We denote random variables with capital letters
- Like variables in a CSP, random variables have domains
 - R in $\{\text{true}, \text{false}\}$ (often write as $\{+r, -r\}$)
 - T in $\{\text{hot}, \text{cold}\}$
 - D in $[0, \infty)$
 - L in possible locations, maybe $\{(0,0), (0,1), \dots\}$



Probability Distributions

- Associate a probability with each value

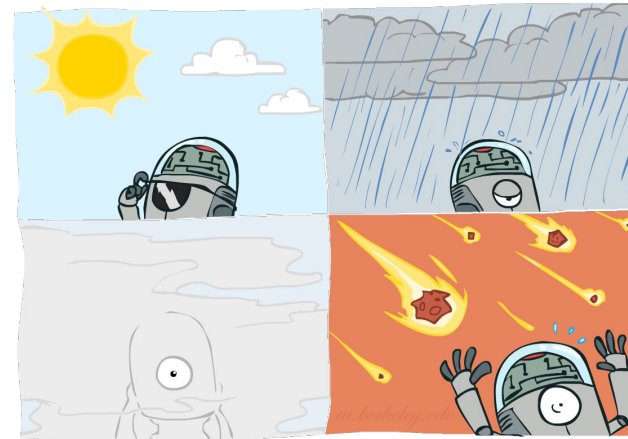
- Temperature:



$P(T)$

T	P
hot	0.5
cold	0.5

- Weather:



$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Probability Distributions

- Unobserved random variables have distributions

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Shorthand notation:

$$P(\text{hot}) = P(T = \text{hot}),$$

$$P(\text{cold}) = P(T = \text{cold}),$$

$$P(\text{rain}) = P(W = \text{rain}),$$

...

OK *if* all domain entries are unique

- A distribution is a TABLE of probabilities of values
- A probability (lower case value) is a single number

$$P(W = \text{rain}) = 0.1$$

- Must have: $\forall x \ P(X = x) \geq 0$ and $\sum_x P(X = x) = 1$

Joint Distributions

- A *joint distribution* over a set of random variables: $X_1, X_2, \dots, X_n$ specifies a real number for each assignment (or *outcome*):

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

$$P(x_1, x_2, \dots, x_n)$$

- Must obey: $P(x_1, x_2, \dots, x_n) \geq 0$

$$\sum_{(x_1, x_2, \dots, x_n)} P(x_1, x_2, \dots, x_n) = 1$$

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

- Size of distribution if n variables with domain sizes d ?
 - For all but the smallest distributions, impractical to write out!

Probabilistic Models

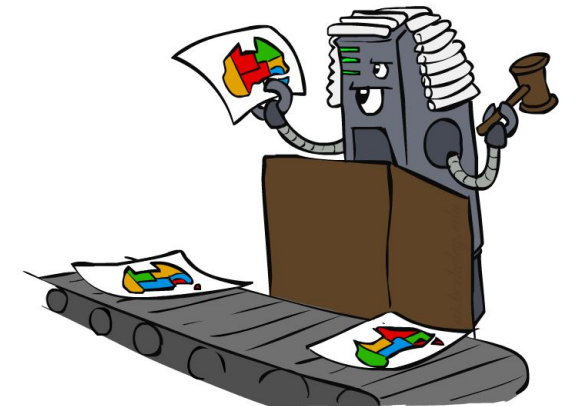
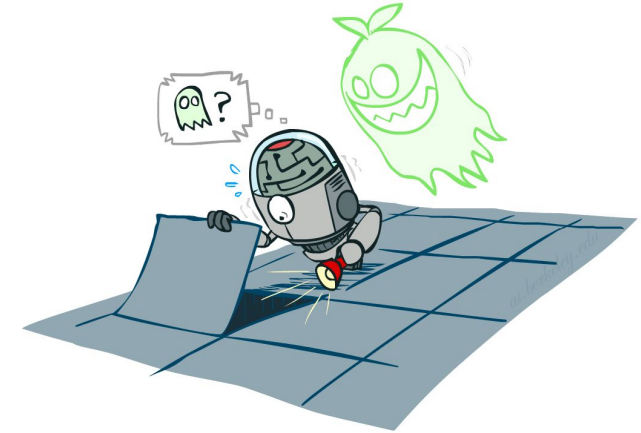
- A probabilistic model is a joint distribution over a set of random variables
- Probabilistic models:
 - (Random) variables with domains
 - Assignments are called *outcomes*
 - Joint distributions: say whether assignments (outcomes) are likely
 - *Normalized*: sum to 1.0
 - Ideally: only certain variables directly interact
- Constraint satisfaction problems:
 - Variables with domains
 - Constraints: state whether assignments are possible
 - Ideally: only certain variables directly interact

Distribution over T,W

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Constraint over T,W

T	W	P
hot	sun	T
hot	rain	F
cold	sun	F
cold	rain	T



Events

- An *event* is a set E of outcomes

$$P(E) = \sum_{(x_1 \dots x_n) \in E} P(x_1 \dots x_n)$$

- From a joint distribution, we can calculate the probability of any event
 - Probability that it's hot AND sunny?
 - Probability that it's hot?
 - Probability that it's hot OR sunny?
- Typically, the events we care about are *partial assignments*, like $P(T=\text{hot})$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Quiz: Events

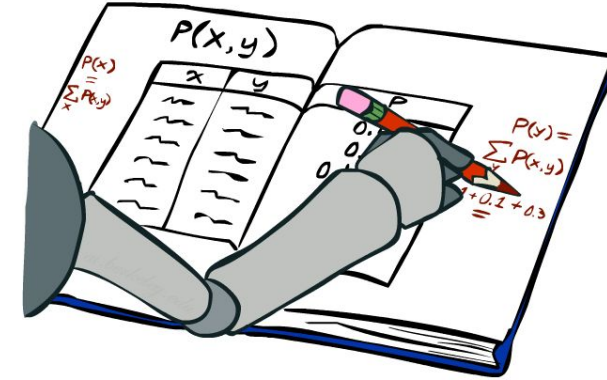
- $P(+x, +y)$?
- $P(+x)$?
- $P(-y \text{ OR } +x)$?

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

Marginal Distributions

- Marginal distributions are sub-tables which eliminate variables
- Marginalization (summing out): Combine collapsed rows by adding



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3



$$P(t) = \sum_s P(t, s)$$

$P(T)$

T	P
hot	0.5
cold	0.5



$$P(s) = \sum_t P(t, s)$$

$P(W)$

W	P
sun	0.6
rain	0.4

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

Quiz: Marginal Distributions

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1



$$P(x) = \sum_y P(x, y)$$

$P(X)$

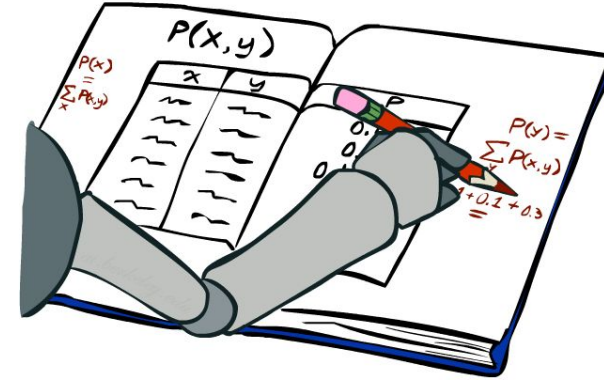
X	P
+x	
-x	

$P(Y)$

Y	P
+y	
-y	



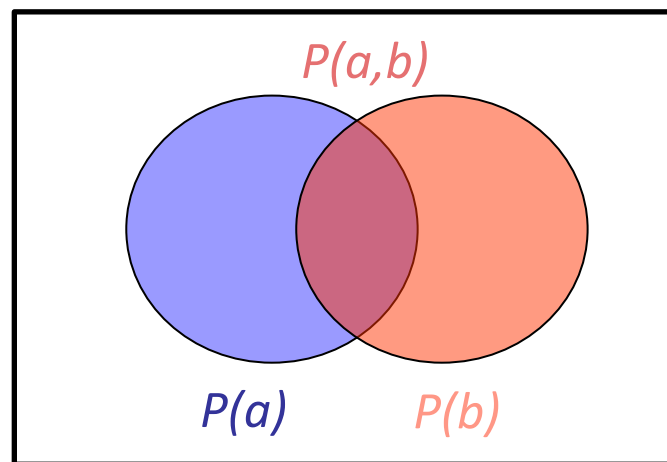
$$P(y) = \sum_x P(x, y)$$



Conditional Probabilities

- A simple relation between joint and conditional probabilities
 - In fact, this is taken as the *definition* of a conditional probability

$$P(a|b) = \frac{P(a, b)}{P(b)}$$



$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(W = s|T = c) = \frac{P(W = s, T = c)}{P(T = c)} = \frac{0.2}{0.5} = 0.4$$

$$\begin{aligned} &= P(W = s, T = c) + P(W = r, T = c) \\ &= 0.2 + 0.3 = 0.5 \end{aligned}$$

Quiz: Conditional Probabilities

▪ $P(+x \mid +y)$?

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

▪ $P(-x \mid +y)$?

▪ $P(-y \mid +x)$?

Conditional Distributions

- Conditional distributions are probability distributions over some variables given fixed values of others

Conditional Distributions

$P(W T)$	$P(W T = \text{hot})$	
	W	P
	sun	0.8
	rain	0.2
	$P(W T = \text{cold})$	
	W	P
	sun	0.4
	rain	0.6

Joint Distribution

$P(T, W)$		
T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Conditional Distributions

- Conditional distributions are probability distributions over some variables given fixed values of others

Conditional Distributions

Joint Distribution

$P(W|T)$

$P(W T = \text{hot})$	
W	P
sun	0.8
rain	0.2

$P(W T = \text{cold})$	
W	P
sun	0.4
rain	0.6

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Probabilistic Inference

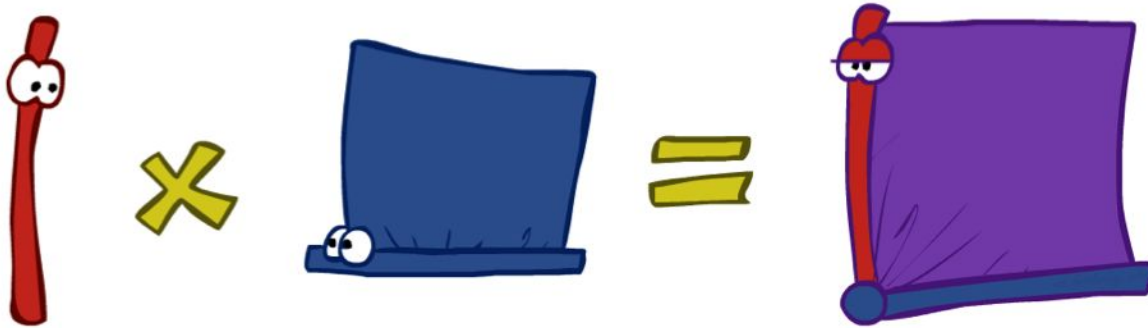
- Probabilistic inference: compute a desired probability from other known probabilities (e.g. conditional from joint)
- We generally compute conditional probabilities
 - $P(\text{on time} \mid \text{no reported accidents}) = 0.90$
 - These represent the agent's *beliefs* given the evidence
- Probabilities change with new evidence:
 - $P(\text{on time} \mid \text{no accidents, 5 a.m.}) = 0.95$
 - $P(\text{on time} \mid \text{no accidents, 5 a.m., raining}) = 0.80$
 - Observing new evidence causes *beliefs to be updated*



The Product Rule

- Sometimes have conditional distributions but want the joint

$$P(y)P(x|y) = P(x, y) \quad \longleftrightarrow \quad P(x|y) = \frac{P(x, y)}{P(y)}$$



The Product Rule

$$P(y)P(x|y) = P(x, y)$$

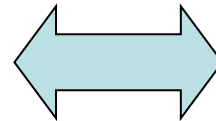
- Example:

$P(W)$

R	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3



$P(D, W)$

D	W	P
wet	sun	
dry	sun	
wet	rain	
dry	rain	

The Chain Rule

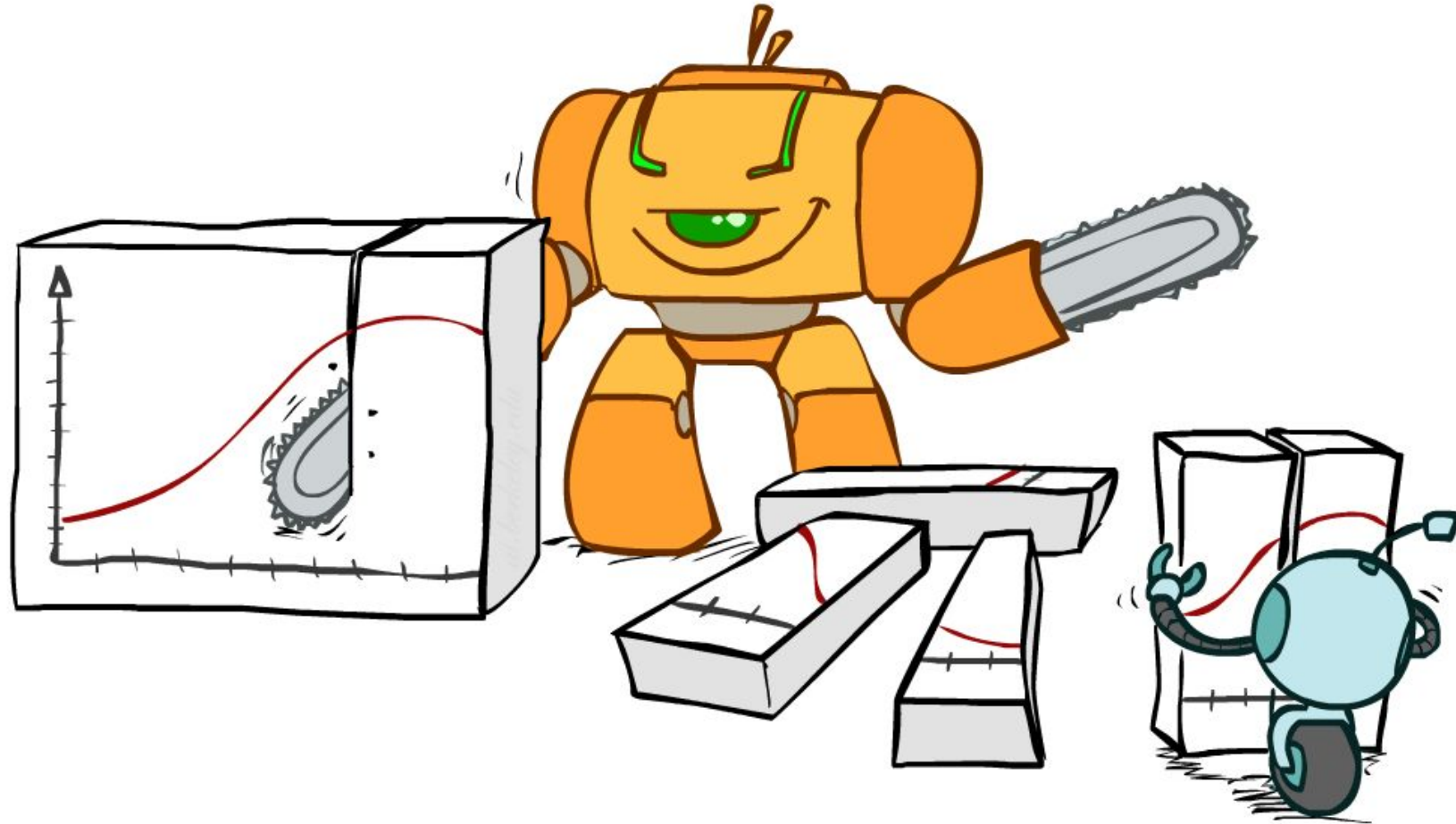
- More generally, can always write any joint distribution as an incremental product of conditional distributions

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \dots, x_n) = \prod_i P(x_i|x_1 \dots x_{i-1})$$

- Why is this always true?

Bayes Rule



Bayes' Rule

- Two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

That's my rule!

- Dividing, we get:

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

- Why is this at all helpful?
 - Lets us build one conditional from its reverse
 - Often one conditional is tricky but the other one is simple
 - Foundation of many systems we'll see later (e.g. ASR, MT)
- In the running for most important AI equation!



Inference with Bayes' Rule

- Example: Diagnostic probability from causal probability:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

- Example:

- M: meningitis, S: stiff neck

$$\left. \begin{aligned} P(+m) &= 0.0001 \\ P(+s|+m) &= 0.8 \\ P(+s|-m) &= 0.01 \end{aligned} \right\} \text{Example gives}$$

$$P(+m|+s) = \frac{P(+s|+m)P(+m)}{P(+s)} = \frac{P(+s|+m)P(+m)}{P(+s|+m)P(+m) + P(+s|-m)P(-m)} = \frac{0.8 \times 0.0001}{0.8 \times 0.0001 + 0.01 \times 0.999}$$

- Note: posterior probability of meningitis still very small
- Note: you should still get stiff necks checked out! Why?

Quiz: Bayes' Rule

- Given:

$$P(W)$$

R	P
sun	0.8
rain	0.2

$$P(D|W)$$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

- What is $P(W \mid \text{dry})$?

Quiz: Bayes' Rule

Q: A doctor knows that the disease **meningitis** causes a patient to have a **stiff neck**, say, 70% of the time. The doctor also knows some unconditional facts: the prior probability that any patient has **meningitis** is $1/50,000$, and the prior probability that any patient has a stiff neck is 1%. Letting s be the proposition that the patient has a stiff neck and m be the proposition that the patient has meningitis, What is the probability that a patient who has a stiff neck actually has meningitis?

Quiz: Bayes' Rule

Q: A doctor knows that the disease **meningitis** causes a patient to have a **stiff neck**, say, 70% of the time. The doctor also knows some unconditional facts: the prior probability that any patient has **meningitis** is 1/50,000, and the prior probability that any patient has a stiff neck is 1%. Letting s be the proposition that the patient has a stiff neck and m be the proposition that the patient has meningitis, What is the probability that a patient who has a stiff neck actually has meningitis?

$$P(s|m) = 0.7$$

$$P(m) = 1/50000$$

$$P(s) = 0.01$$

$$P(m|s) = \frac{P(s|m)P(m)}{P(s)} = \frac{0.7 \times 1/50000}{0.01} = 0.0014.$$

Independence

- Two variables are *independent* in a joint distribution if:

$$P(X, Y) = P(X)P(Y)$$

$$X \perp\!\!\!\perp Y$$

$$\forall x, y \ P(x, y) = P(x)P(y)$$

- Says the joint distribution *factors* into a product of two simple ones
 - Usually variables aren't independent!
- Can use independence as a *modeling assumption*
 - Independence can be a simplifying assumption
 - Empirical* joint distributions: at best “close” to independent
 - What could we assume for {Weather, Traffic, Cavity}?
- Independence is like something from CSPs: what?



Independence

- $P(\text{toothache}, \text{catch}, \text{cavity}, \text{cloud}) = P(\text{cloud} | \text{toothache}, \text{catch}, \text{cavity})P(\text{toothache}, \text{catch}, \text{cavity})$.
- $P(\text{toothache}, \text{catch}, \text{cavity}, \text{cloud}) = P(\text{cloud})P(\text{toothache}, \text{catch}, \text{cavity})$.

$$P(X|Y) = P(X) \quad \text{or} \quad P(Y|X) = P(Y) \quad \text{or} \quad P(X,Y) = P(X)P(Y).$$



Example: Independence?

$P_1(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$P(T)$

T	P
hot	0.5
cold	0.5

$$P_2(T, W) = P(T)P(W)$$

T	W	P
hot	sun	0.3
hot	rain	0.2
cold	sun	0.3
cold	rain	0.2

$P(W)$

W	P
sun	0.6
rain	0.4

Example: Independence

- N fair, independent coin flips:

$P(X_1)$

H	0.5
T	0.5

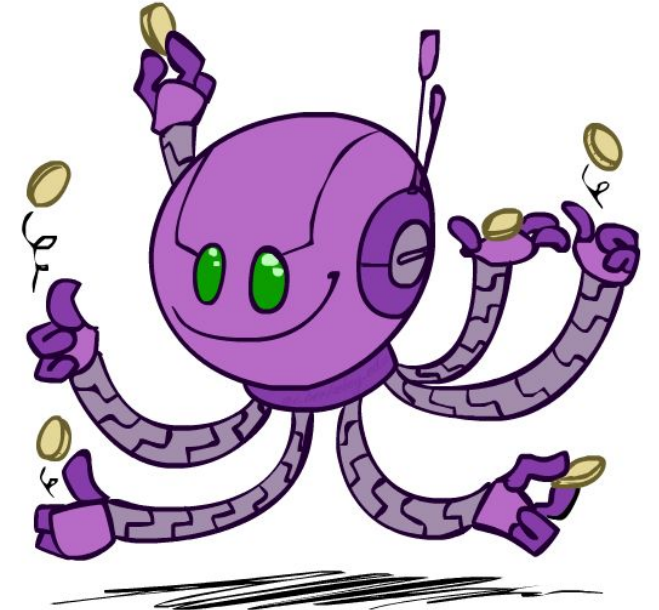
$P(X_2)$

H	0.5
T	0.5

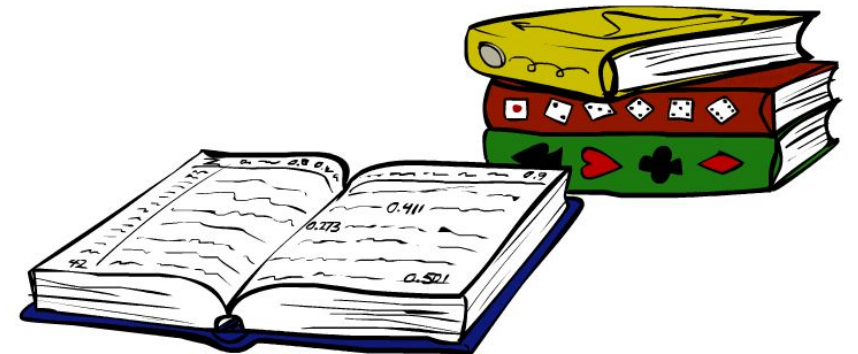
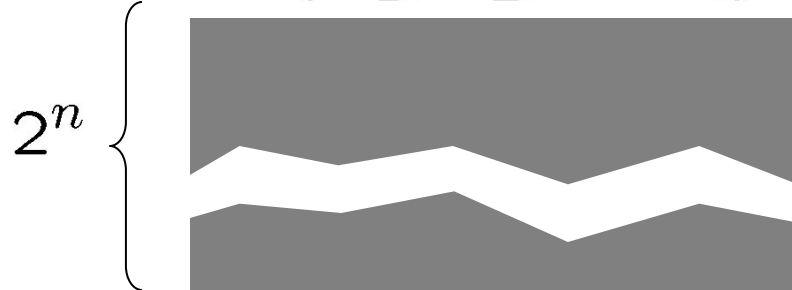
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$P(X_n)$

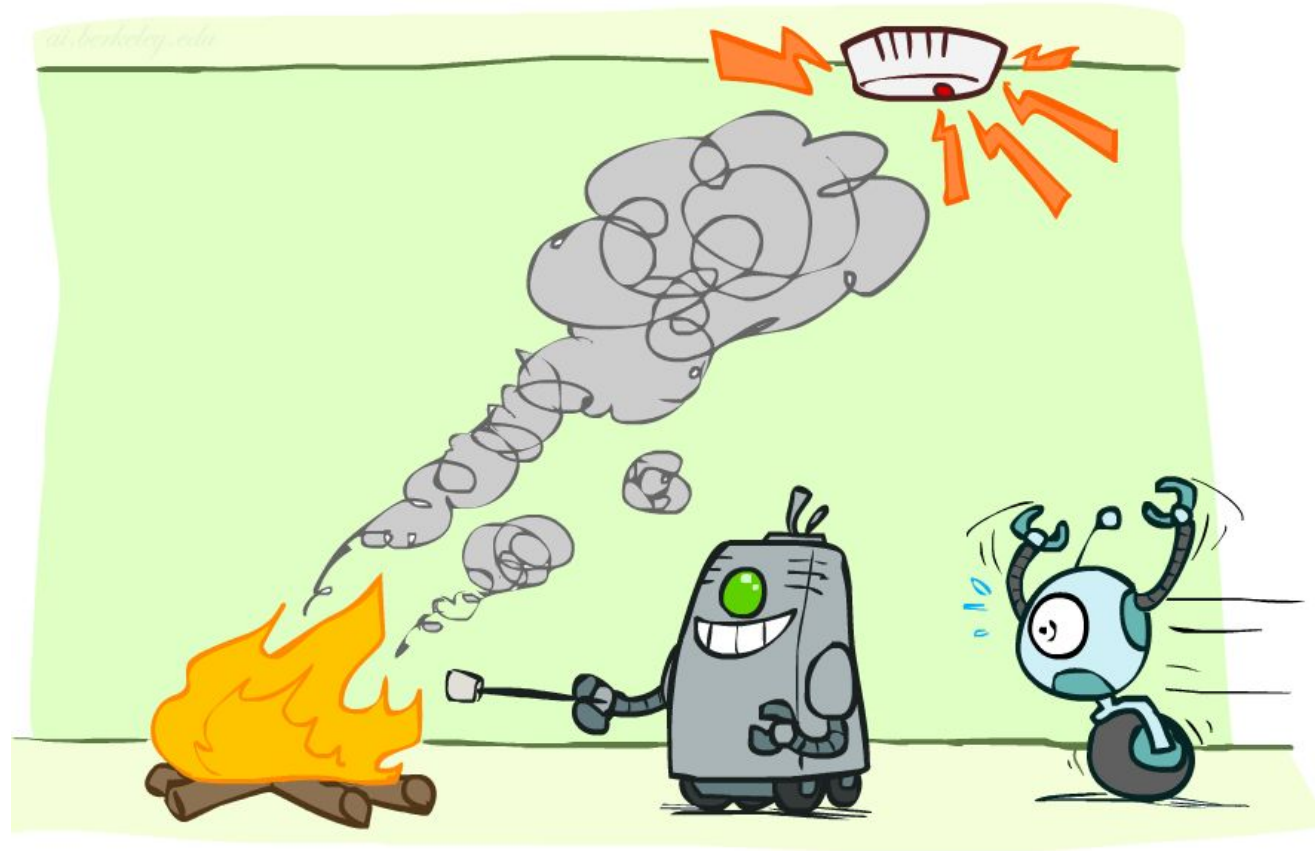
H	0.5
T	0.5



$P(X_1, X_2, \dots, X_n)$



Conditional Independence



Conditional Independence

It refers to a situation where two random variables, X and Y, are independent given a third variable Z. This means that once we know Z, knowing X gives us no additional information about Y and vice versa.

Two random variables X and Y are **conditionally independent** given a third variable Z if:

$$P(X, Y | Z) = P(X | Z)P(Y | Z)$$

or equivalently,

$$P(X | Y, Z) = P(X | Z)$$

Conditional Independence

- $P(\text{Toothache}, \text{Cavity}, \text{Catch})$
- If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache:
 - $P(+\text{catch} \mid +\text{toothache}, +\text{cavity}) = P(+\text{catch} \mid +\text{cavity})$
- The same independence holds if I don't have a cavity:
 - $P(+\text{catch} \mid +\text{toothache}, -\text{cavity}) = P(+\text{catch} \mid -\text{cavity})$
- Catch is *conditionally independent* of Toothache given Cavity:
 - $P(\text{Catch} \mid \text{Toothache}, \text{Cavity}) = P(\text{Catch} \mid \text{Cavity})$
- Equivalent statements:
 - $P(\text{Toothache} \mid \text{Catch}, \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity})$
 - $P(\text{Toothache}, \text{Catch} \mid \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity}) P(\text{Catch} \mid \text{Cavity})$
 - One can be derived from the other easily

